

Chapter 31 — Tuberculosis and Parasitic Infections of the Genitourinary Tract

UROGENITAL TUBERCULOSIS

- **Tuberculosis (TB) can affect any organ**, including the urogenital (UG) tract; untreated UG TB causes **irreparable damage — renal failure and infertility** — so TB belongs in the differential of UG disorders.
- Called the **second “great imitator” (after syphilis)**: it mimics many diseases and delays correct diagnosis.
- As TB becomes rarer in industrialised nations, diagnosis increasingly depends on **clinical recognition and a high index of suspicion**.
- **“Urogenital TB” is now preferred to “genitourinary TB”** because the urinary tract is affected far more often than the genital tract.

History

- ***Mycobacterium tuberculosis* coevolved with humans**; its progenitor (*M. prototuberculosis*) may have infected early hominids >3 million years ago. Evidence runs from a *Homo erectus* skeleton, through Neolithic DNA, to Egyptian mummies (c. 3000 BCE).
- TB reached **epidemic proportions in the 18th–19th centuries** in Europe and North America (“consumption”, up to a quarter of deaths in the Industrial Age).
- **24 March 1882: Robert Koch** presented the first isolation of the tubercle bacillus. This date is now **World Tuberculosis Day**.

Microbiology

- TB is caused by the acid-fast ***Mycobacterium tuberculosis* complex (MTBC)**:

Host range	Species
Humans only	<i>M. tuberculosis</i> , <i>M. africanum</i>
Humans and other mammals	<i>M. bovis</i> , <i>M. canettii</i> , <i>M. microti</i> , <i>M. caprae</i> , <i>M. orygis</i> , <i>M. pinnipedii</i>
Animals only (so far)	<i>M. mungi</i> , <i>M. suricattae</i>

- **Commonest human isolate: *M. tuberculosis*, then *M. bovis***. There is local variation: *M. africanum* causes up to 40% of TB in West Africa.
- “*M. tuberculosis*” is often used loosely for the whole complex. The human-infecting species cause **clinically indistinguishable disease but differ in drug susceptibility**: *M. bovis* is innately resistant to pyrazinamide.

Epidemiology

- The World Health Organization (WHO) estimates that **one-quarter of the world’s population has latent MTBC infection**. In 2021 there were 10.6 million new active cases and 1.6 million deaths; mortality has fallen 45% since 1990.
- **Obstacles to control**: human immunodeficiency virus (HIV) in sub-Saharan Africa, rising obesity and diabetes, and multidrug and extensive drug resistance.
- **United States**: about 2.5 cases per 100,000 (2022). Incidence peaked in 1992 and has declined since. There was a COVID-19 dip in 2020, back to prepandemic levels by 2022. Incidence is **16-fold higher in non-US-born persons**.

- **UG involvement: 15–20% of pulmonary TB patients in developing countries vs 2–10% in developed countries;** >90% of UG TB occurs in developing countries.
- **Rank among extrapulmonary sites: second after lymph nodes** in the developing world; **third after pleural and lymphatic** in the US (27% of extrapulmonary cases).
- UG TB is **about two-thirds male** and mainly a **disease of adults**, although it has been reported from age 2.

Infection, Host Immune Response, and Transmission

- **Entry by inhalation of cough-generated aerosols;** direct soft-tissue inoculation is rare. The course then runs: **alveolar macrophage phagocytosis** → either killed and cleared (no infection, no adaptive response) or **replication inside macrophages** → spread via lymphatics to **hilar nodes** → **haematogenous seeding of distant organs**. This can happen during the **up to 12 weeks before cellular immunity is detectable**.
- **Granuloma formation:** infected macrophages secrete interleukin-6 (IL-6), IL-12, IL-1 β and tumour necrosis factor- α (TNF- α). Foamy macrophages, epithelioid cells and **multinucleated giant cells (Langhans cells)** form the centre, with a lymphocyte cuff.
- **T cells secrete IL-2, TNF- α and, above all, interferon- γ (IFN- γ),** which maintains the granuloma and drives macrophage killing. Where killing fails, the granuloma **sequesters dormant bacilli**.

Outcome after infection	Proportion / risk	Features
Controlled ("latent")	90–95%	Cicatrization, granuloma calcification
Primary progression	<5%	Active disease within the first year
Reactivation	Lifetime risk 5–15%; higher with comorbidity	Triggered by old age, renal failure, diabetes, malnutrition, HIV, immunosuppression → renewed replication and caseation (the characteristic lesion)

- **Exclusively extrapulmonary disease (e.g., UG TB) is generally not infectious.** Exceptions are direct skin inoculation and aerosol-generating procedures on infected fluids and tissues.
- **Latent vs active is now seen as a spectrum** (contained infection → subclinical disease → infectious disease). Patients move in both directions as host immunity or treatment changes.

Development of Urogenital Disease

Route	Frequency	Key points
Haematogenous	Principal route	Seeds kidneys and epididymis ; other UG organs are infected by contiguous spread from these. Usually prolonged latency (up to 46 years) before clinical disease.
Ascending / retrograde	Much less common	Classic setting is intravesical bacille Calmette-Guérin (BCG) , a live attenuated <i>M. bovis</i> vaccine, which complicates about 0.9% of instillations. Causes pyelonephritis, renal abscess, ureteric obstruction, cystitis, prostatitis and epididymo-orchitis.
Contiguous spread	Rare	TB "does not respect anatomical boundaries": spine → psoas → kidney; gastrointestinal (GI) TB → enterorenal or enterovesical fistulae.
Direct inoculation	Exceedingly rare	Autoinoculation of the genitalia from infected stool or urine; genital or oral contact with infected lesions.

Clinical Manifestations and Pathological Features

- **Symptoms are nonspecific.** Patients are often treated repeatedly for bacterial infection, or investigated for malignancy, before TB is considered. **Renal TB can be silent until it reaches the bladder.**

- **Constitutional symptoms** (fever, weight loss, night sweats, malaise) occur in **<20%**. Presentations: dysuria alone up to 50%, storage symptoms 50%, haematuria and flank pain 33%. **Renal colic <10%**, from passage of necrotic papillae, clots, stones or caseous material. A minority are asymptomatic.
- **Hallmark laboratory finding: sterile pyuria and/or haematuria** (>90% in developing countries).

Kidney

- **80% of UG TB is renal**; infection is progressive and destructive.
- **Collateral renal damage without imaging change**: in pulmonary TB with renal failure, biopsy may show **TB granulomatous interstitial nephritis**, which can reverse with treatment. Immune-complex glomerulonephritis and secondary amyloidosis also occur.
- **Miliary TB**: massive haematogenous seeding gives innumerable ~3-mm pale granulomas (“millet seeds”) through cortex and medulla. It carries **high mortality** but **usually spares renal function**.
- **Localised renal TB** progresses as: bacilli lodge in **periglomerular capillaries** → parenchymal granulomas coalesce → caseation → cavities → abscess, chronic pyelonephritis, parenchymal and papillary necrosis, flank sinus tracts. Costovertebral angle tenderness may be present. Calyces become inflamed and then calcify → **distortion, dilatation, stenosis**.
- **Autonephrectomy** (nonfunctioning kidney) occurs in **up to 33%**:

Caseo-cavernous autonephrectomy	Fibrotic autonephrectomy
Viable tissue replaced by granulomas and exudate-filled cavities; with or without calcification	Severe scarring and calcification → small shrunken kidney

- **End-stage renal failure in about 7%**. Chronic inflammation can cause **squamous metaplasia of the renal pelvis** that persists after treatment, a risk for **squamous cell carcinoma (SCC)**.

Ureter

- **Always descending from the kidney**. Mural granulomas and descending infected calculi cause scarring and strictures, **commonest distally at the vesicoureteric junction**.
- **Pan-ureteric strictures** give a “**beaded corkscrew**” ureter. Distortion causes both **obstruction and reflux**, and obstruction is an **important cause of renal failure**. Spontaneous ureteric perforation above a stricture has been reported once.

Bladder

- Disease **begins near the ureteric orifices** and spreads by lymphatics as a patchy cystitis, with ulceration where granulomas coalesce. The **dome is most affected; trigone and neck usually spared**.
- **Contracture after about 1 year of chronic inflammation**. Frequency, urgency, pain and dysuria become prominent at **capacity <100 mL**; the “**thimble**” bladder is **<20 mL**. It is a **late complication**, commoner in developing countries (diagnosed later). Rarely, **spontaneous perforation presents as an acute abdomen**, often recognised only at laparotomy.

Epididymis, Vas Deferens, Testes, and Scrotum

- **Epididymis is the second commonest UG site of haematogenous seeding** after the kidney. It is involved in 10–55% of UG TB and is **bilateral in about one-third**.
- Disease **starts in the more vascular globus minor (tail)** → fibrous luminal obliteration → nodular, swollen or hard epididymis. **A posterior scrotal sinus tract develops in up to 50%**. Spread to the vas gives a **thickened “beaded” vas**.
- **Isolated testicular or epididymal TB is rare**; the testis is usually involved by extension from the epididymis. Hard masses **mimic testicular tumour**. **Hydrocele in about 5%**.

Prostate and Seminal Vesicles

- **Prostate involved in 22–49%**, by one of two routes:

Haematogenous prostatic TB	Urinary-route prostatic TB
Peripheral lesions, urethra spared ; asymptomatic → calcification and hardening	Involves the urethra ; behaves like bacterial prostatitis

- **Suspect TB in chronic prostatitis that persists despite antibiotics.** Quinolones are active against MTBC, but **short prostatitis courses are insufficient**, so symptoms fail to resolve or quickly recur. Prostatic abscess is rare, mainly seen in acquired immunodeficiency syndrome (AIDS).
- **Seminal vesicle TB can present as infertility**, sometimes the first symptom. Spread is via the vas (from testis or epididymis) or via the urethra and ejaculatory ducts (from kidney, bladder or prostate). Features: low-volume ejaculate, oligo- or azoospermia, haemospermia. Vesicles are enlarged early and become hard nodules late; abscess is rare.

Urethra and Penis

- **Urethra is relatively resistant** (1.9–4.5%). It is usually involved with prostatic TB and can present with **urethroscrotal fistulae**; isolated urethral TB is very rare.
- **Primary penile TB is exceedingly rare.** It follows a sequence: inflamed papule or keratotic plaque (**lupus vulgaris**) → ulceration → cavernosal spread → pea-sized nodules (painless and hard, or fungating). It **mimics malignancy**; fibrosis causes deformity and erectile dysfunction.

	Orificial TB	Papulonecrotic tuberculid (PNT)
Nature	Rapidly necrotic penile TB from autoinoculation (stool or urine); rarely haematogenous or lymphatic	Hypersensitivity reaction to disseminated MTBC antigens from a distant focus
Host	Immunocompromised or debilitated; marks very advanced UG or GI TB	Described in Japan, South Africa, India; glans and other skin
Lesion	Painful ulcers with pseudomembrane, eroding deeply	Red papules → ulcers (can be painless) → varioliform scars
Bacilli	Present	Absent: culture- and usually polymerase chain reaction (PCR)-negative ; histology often inconclusive
Diagnosis / outcome	Poor prognosis	Mimics syphilis, Behçet disease, herpes, balanitis, SCC; confirmed by response to empirical TB treatment

Concept — Every UG organ follows the same arc (*granuloma* → *caseation* → *fibrosis and calcification*), so early disease mimics infection or tumour and late disease presents as obstruction, contraction or infertility. The damage is **cicatricial, and much of it outlasts the infection**.

Diagnosis

- **Primary goal: isolate MTBC in culture for drug susceptibility testing.** If cultures and DNA tests are negative, **caseating granulomas on tissue** support the diagnosis. Failing both, diagnosis rests on consistent findings, probable exposure and **response to empirical treatment**.
- **Up to 20% of UG TB coexists with pulmonary TB**, so always look for pulmonary disease.

Culture

- **Gold standard: urine acid-fast bacilli (AFB) culture.** Collect **first-void urine** (most concentrated) on **3–5 consecutive days** and **culture immediately**, because urinary acidity retards growth. Sensitivity is up to 80–90% when done this way but as low as 10% in real-world practice (sporadic shedding).

- **Ziehl-Neelsen staining of urine: sensitivity <50%.** Always culture any biopsy or surgical tissue.

MYCOBACTERIAL CULTURE SYSTEMS

Medium	Type	Time to detection	Setting
Löwenstein-Jensen (LJ)	Solid, egg-based	4–6 weeks; laborious	Developing countries: cheapest, no special equipment
Middlebrook 7H10	Solid, agar-based, transparent	About 1 week earlier than LJ	Developed countries
BACTEC Mycobacteria Growth Indicator Tube (MGIT)	Liquid, automated fluorescence quenching	As little as 10 days	Developed countries; in-house first-line susceptibility testing

- **Guidelines: culture on at least one solid medium alongside the liquid system.** Semiautomated radiometric liquid culture is also available.
- **Second-line susceptibility and molecular resistance sequencing** are generally available only at reference laboratories.

Nucleic Acid Amplification Tests

- **Nucleic acid amplification tests (NAATs) give results in 1–2 days** and help when bacillary load is low. **Urine contains amplification inhibitors that lower sensitivity.** Performance depends on specimen, target sequence and protocol.
- **GeneXpert MTB/RIF (since 2010):** self-contained and automated; detects MTBC and **rifampicin resistance** simultaneously. Because >90% of rifampicin-resistant strains are also isoniazid (INH)-resistant, **rifampicin resistance is a surrogate for multidrug-resistant TB (MDR-TB)** (resistance to INH and rifampicin).
- Systematic reviews in urinary TB show NAAT sensitivity 70–100% (small, heterogeneous studies). Pooled GeneXpert values in urine against AFB culture: **sensitivity 83–95%, specificity 79–99%.** NAAT also performs well on epididymal tissue.
- **US position: NAAT is an adjunct and does not replace culture, and must not be used to exclude extrapulmonary TB.** WHO recommends GeneXpert in the initial work-up of all TB, although cost limits its use in developing countries.
- **NAAT cannot monitor treatment:** nucleic acid from dead organisms keeps the test positive.

Urine Lipoarabinomannan Assay

- **Lipoarabinomannan (LAM)** is a cell-wall glycolipid found in urine, sputum and serum. The **lateral flow urine assay (LF-LAM)** is point-of-care, needs no laboratory equipment and gives a result in <30 minutes.
- In persons living with HIV (PLHIV), **sensitivity is low but specificity high** (meta-analysis). WHO (2019) recommends it for **PLHIV with TB symptoms, advanced HIV illness, or CD4 <200 cells/mm³.** In the US its use is mainly academic.

Histopathology

- Used when urine cultures are negative. **Caseating granulomas in the right clinical context** establish the diagnosis even if mycobacteria are not seen. Histology is consistent with TB in only a minority of UG TB cases.
- **Fine-needle aspiration cytology of epididymal nodules** is diagnostic in about two-thirds. **Adding NAAT to tissue increases sensitivity.**

Screening Tests

- **Tuberculin skin test (TST) and IFN-γ release assays (IGRAs) cannot distinguish latent infection from active disease:** a positive test does not rule TB in, and a negative test does not rule it out. Their role is screening for infection.

- **IGRA is preferred for persons ≥ 5 years old born in endemic countries**; TST is an acceptable alternative.

Tuberculin skin test (purified protein derivative, Mantoux test). Tuberculin is injected **intradermally into the volar forearm** and read at **48–72 hours**, when delayed-type hypersensitivity induration appears in primed individuals. Centers for Disease Control and Prevention (CDC) cut-offs are risk-stratified:

TST POSITIVITY THRESHOLDS (INDURATION)

Induration	Positive in
≥ 5 mm	Recent TB contact; fibrotic chest radiograph consistent with prior TB; immunosuppressed
≥ 10 mm	Immigrants from high-prevalence countries; residents or workers in high-risk institutions; injection drug users; comorbidities raising TB risk
≥ 15 mm	General population (no risk factors)

- **TST pros:** cheap, no laboratory, easy (after training). **Cons: not specific** (BCG, nontuberculous mycobacteria), needs a **second visit**, and gives **false negatives in 10–25% of active TB**. It is positive in most UG TB.

Interferon- γ release assays. These blood tests measure IFN- γ released on exposure to MTBC-specific antigens, effectively an in vitro TST. Antigens are **absent from BCG and most nontuberculous mycobacteria**. Result in 24 hours.

COMPARISON: TST VS THE TWO IGRAS

	TST	QuantiFERON-TB Gold Plus (QFT-Plus)	T-SPOT.TB
Sample	Intradermal injection	Whole blood in 4 tubes : ESAT-6/CFP-10; CD8-activating antigens; negative and positive controls	Separated peripheral blood mononuclear cells (PBMCs) + ESAT-6/CFP-10
Read-out	Induration at 48–72 h	Incubate 16–24 h $\rightarrow$ enzyme-linked immunosorbent assay (ELISA) of plasma IFN- γ	Enzyme-linked immunospot (ELISPOT) ; spots counted manually
Results	Positive / negative by cut-off	Positive / negative / indeterminate	Positive / negative / borderline
BCG cross-reaction	Yes	No	No
Sensitivity (culture-confirmed TB, pooled)	89%	83% (third generation; fourth-generation data pending)	91% — most sensitive, but harder to perform

- The **fourth-generation QFT-Plus** was introduced in 2017.

Radiography

- The **choice of test depends on disease location and symptoms**. Imaging is often the first clue that TB is the cause.

Plain radiography.

- The **kidney-ureter-bladder (KUB) film shows calcification in 24–44%** (Fig. 31.1). Progression runs: faint punctate parenchymal calcification $\rightarrow$ **globular calcification** (tuberculous mass) $\rightarrow$ **triangular ring-like calcification** (papillary necrosis) $\rightarrow$ **“cement” or “putty” kidney** in fibrotic autonephrectomy. Calcific rims outlining each lobe form a **lobar pattern that is pathognomonic of end-stage renal TB**.
- Infected calculi form odd shapes in a deformed pelvis; an **upward-arrowhead stone suggests a “hiked-up” pelvis** contracted by scarring.

- **Ureteric calcification in TB is intraluminal** (vs mural in schistosomiasis). Bladder wall calcification is uncommon except in late contraction. **Prostate and seminal vesicle calcification in about 10%.** Look for **vertebral erosion and psoas cold-abscess calcification.**
- **Every patient evaluated for UG TB needs a chest radiograph** to exclude infectious pulmonary TB.

Intravenous urography.

- Intravenous urography (IVU) was **formerly the gold standard** for early renal TB and is still preferred first-line in some regions. It has largely been **replaced by computed tomography (CT), CT urography, magnetic resonance imaging (MRI) and ultrasound (US).**
- **Signs of early disease:** loss of calyceal sharpness and edge irregularity; **“moth-eaten” calyces**; filling defects (tuberculoma rupture, papillary necrosis); medullary cavities communicating with the collecting system; the **“phantom calyx”**, where a stenosed calyx or infundibulum fails to opacify.
- **Ureter:** a rigid, straight, aperistaltic **“pipestem” ureter** or a **“beaded corkscrew” ureter**. Both are highly suggestive of TB, especially with renal or bladder change. IVU also shows nonfunction (autonephrectomy) and a **contracted bladder** (Fig. 31.2). A perinephric abscess is suggested by restricted renal movement with breathing or by ureteric displacement.
- **Commonest IVU findings are obstructive:** calyceal obliteration (Fig. 31.3), infundibular narrowing, hydrocalycosis, segmental or total hydronephrosis, hydroureter (Fig. 31.4). Dilated, distorted calyces form a **cloverleaf pattern**. Vesicoureteric obstruction comes from cystitis or distal-third stricture (Fig. 31.5).
- **Kerr’s kink:** a “hiked-up” renal pelvis with **sharp angulation of the pelviureteric junction (PUJ).**

Computed tomography with urography.

- **The most-used modality in developed countries.** It shows calcification, scarring and obstruction; multidetector scanners detect lesions of **3–4 mm**; contrast phases assess function.
- **Small granulomas are hypoattenuating with little or no enhancement. Large granulomas mimic renal cell carcinoma** (mass-like, peripheral enhancement).
- CT is **more sensitive than KUB for calcification** and collecting-system thickening, and best for **complicated or extensive disease:** perinephric and psoas abscess, nodes, vertebrae, spleen, liver, and prostate or seminal vesicle enlargement, cavitation, abscess and calcification (Fig. 31.6: a right kidney hydronephrotic from infundibular stenosis but still functioning, beside an end-stage atrophic calcified left kidney).
- **Disadvantages: higher radiation dose than IVU; less sensitive than IVU for minimal urothelial thickening and subtle papillary necrosis**, so IVU may still be preferred for early disease in resource-constrained settings. **Advantages of contrast CT:** shows parenchyma, oedema and hypoperfusion.

Retrograde and antegrade pyelography. Largely **replaced by CT urography.** Still useful when **renal insufficiency or contrast allergy precludes IVU or CT**, and to show whether cavities are obstructive and whether they communicate with the collecting system.

Ultrasonography

- **Limited role:** nonspecific findings and poorer definition than CT. Useful in **children and pregnancy** (no radiation) and cheaper.
- Shows scrotal disease (diffuse enlargement, hypoechoic areas, complex collections, hydrocele) and, transrectally, thickened prostate and vesicles. Also shows renal abscesses and cavities: **septated cystic lesions suggest chronic infection**; calcification is echogenic with shadowing. Restricted renal movement suggests perinephric or psoas abscess. Shows ascites, lymphadenopathy and omental caking.
- **Key use: monitoring hydronephrosis during medical treatment**, because healing fibrosis can worsen obstruction.

Magnetic Resonance Imaging

- Used less than contrast CT but useful in **children and pregnancy**. Findings are comparable to CT with **better tissue resolution**; it can detect a single granuloma.
- **Small lesions are hypointense on T1 and T2; larger lesions show central T2 hyperintensity** (central cellularity). Large lesions may be indistinguishable from malignancy.
- **Magnetic resonance urography (MRU) is more sensitive than IVU** for urothelial thickening and caliectasis. **Diffusion-weighted imaging (DWI) separates hydronephrosis from pyonephrosis**; cine and dynamic MRU assess peristalsis. **Apparent diffusion coefficient falls with fibrosis**, which allows staging and monitoring of treatment.
- **Caution with gadolinium in renal failure** (nephrogenic systemic fibrosis).

Cystoscopy and Ureteroscopy

- **Limited diagnostic role**; findings are nonspecific: hyperaemia, erosion, ulceration, granulomatous masses, irregular orifices. **Ulcers can mimic malignancy**.
- A “**golf-hole**” ureteric orifice suggests TB and should prompt upper-tract imaging or endoscopy (Fig. 31.7).
- **Biopsy when possible, especially if malignancy is possible**. Positive urine culture is sufficient but slow. In culture-negative patients, bladder biopsy is 19–52% sensitive.

Fig. 31.7 — *endoscopic spectrum of UG TB*. (A) Extensive renal and ureteric TB with calcification and stricture; (B) acutely inflamed ureteric orifice; (C) tuberculous bullous granulations; (D) acute tuberculous ulcer; (E) golf-hole ureter; (F) golf-hole ureter, severely withdrawn; (G) healed lesion; (H) acute tuberculous cystitis with ulceration.

Concept — Culture is the gold standard because only culture yields susceptibility data and tracks treatment response. Every faster test (NAAT, LAM, histology, immunological assays) trades away one of those properties. Imaging finds the cicatricial pattern, not the organism.

Treatment

- Before antimicrobials, treatment was sanatorium rest and, for severe UG disease, extirpative surgery. **Streptomycin (1944), INH (1952) and the rifamycins (1957)** replaced both.
- Most patients, **even with MDR-TB, are now treated medically as outpatients**. Surgery serves **diagnosis or as an adjunct in advanced disease**.

Medical Therapy

- **Multidrug therapy is mandatory** because:
 - bacilli occupy different microenvironments with different metabolic and replication rates;
 - drugs differ: bactericidal vs bacteriostatic, fast vs slow replicators, tissue penetration and optimal pH;
 - **combinations prevent emergence of resistance**.
- **Start with INH, rifampicin, pyrazinamide and ethambutol**. Baseline tests: platelets, liver and renal function; **HIV test in all**; hepatitis B and C and diabetes screening as appropriate. Tailor to susceptibilities. **Directly observed therapy (DOT)** ensures adherence and limits resistance.

TABLE 31.2 FIRST-LINE ANTITUBERCULOUS DRUGS (ADULT DOSING)

Drug	Daily dose	Intermittent (3×/week, continuation phase only)	Main adverse effects	Key footnote
Isoniazid	5 mg/kg (typically 300 mg)	15 mg/kg (typically 900 mg)	Hepatotoxicity, peripheral neuropathy	Pyridoxine 25–50 mg if malnourished, pregnant, HIV, renal failure, thyroid disease, alcoholism, diabetes
Rifampicin	10 mg/kg (typically 600 mg)	10 mg/kg (typically 600 mg)	Hepatotoxicity, flu-like syndrome, pruritus, drug interactions	Generally not with HIV protease inhibitors or certain non-nucleoside reverse transcriptase inhibitors (NNRTIs)
Rifabutin	5 mg/kg (typically 300 mg)	Not recommended	As rifampicin + uveitis , neutropenia	Dose adjusted with antiretrovirals (raised with efavirenz, reduced with protease inhibitors); watch for uveitis
Rifapentine	—	Once weekly 10–20 mg/kg (≥600 mg), continuation phase only, very select patients	As rifampicin	Contraindicated in HIV, cavitary pulmonary disease and extrapulmonary TB
Pyrazinamide	40–55 kg: 1000 mg; 56–75 kg: 1500 mg; 76–90 kg: 2000 mg	1500 / 2500 / 3000 mg	Arthralgia, hyperuricaemia , hepatotoxicity, pruritus, rash, GI upset	—
Ethambutol	40–55 kg: 800 mg; 56–75 kg: 1200 mg; 76–90 kg: 1600 mg	1200 / 2000 / 2400 mg	Optic neuritis: reduced red-green discrimination and acuity	—

Daily dosing may be 5 days/week DOT; daily is preferred; WHO and CDC no longer recommend dosing less often than 3×/week. From American Thoracic Society, CDC, Infectious Diseases Society of America: Treatment Guidelines, Medical Letter, 2012; American Thoracic Society: Treatment of tuberculosis, MMWR Recomm Rep, 2003.

- **What the table tests:** the signature toxicity of each drug. INH → **neuropathy** (prevented by pyridoxine); ethambutol → **eye**; pyrazinamide → **joints and urate**; rifamycins → **interactions**.
- **Trap:** all except ethambutol are hepatotoxic, and rifapentine is barred from extrapulmonary (hence UG) TB.
- **Second-line agents** are for intolerance, resistance or failure. They vary in tolerability and route:

NEWER SECOND-LINE ANTITUBERCULOUS AGENTS

Agent	Origin / status	Role
Fluoroquinolones	Developed for other bacterial infections	Second-line; resistance defines pre-XDR and XDR-TB
Linezolid	Developed for other bacterial infections	Component of BPAL
Bedaquiline	First new drug in 40 years developed specifically for TB	Component of BPAL
Pretomanid	FDA-approved 2019	With bedaquiline and linezolid for MDR- or XDR-TB (BPAL)

STANDARD SHORT-COURSE REGIMEN FOR UNCOMPLICATED UG TB (6 MONTHS)

Phase	Duration	Drugs	Target
Intensive	2 months, daily	INH + rifampicin + pyrazinamide (+ ethambutol until susceptibility known)	Rapidly multiplying bacilli
Continuation	4 months, daily or 3×/week (not twice weekly)	INH + rifampicin	Slow, sporadic multipliers and persisters

- **Pyridoxine (vitamin B6)** prevents INH neuropathy. **Stop ethambutol once the strain is susceptible** to the other first-line drugs. First-line drugs reach **high urinary concentrations and work well at acid pH**.
- **The 4-month rifapentine–INH–pyrazinamide–moxifloxacin regimen is not recommended for extrapulmonary TB and must not be used for UG TB.**
- **Prolonged courses:** complicated UG TB (abscesses, severe scarring, extensive disease) and its **higher relapse rate than pulmonary TB** lead some clinicians to treat for **12–24 months, or at least 6 months after surgery**. Discuss any deviation from standard therapy with TB specialists.
- **Monitoring:**
 - **monthly liver enzymes if liver disease pre-exists;** hepatotoxicity reverses on stopping, but severe injury has occurred; avoid alcohol and other hepatotoxins;
 - **visual acuity and red-green colour vision on ethambutol;**
 - **renal lesions may worsen on treatment:** healing fibrosis can worsen obstruction and bladder contraction. This may need steroids or surgery for new or worsening obstruction.

Corticosteroids

- The rationale is to **limit immune-mediated tissue destruction and scarring**; their role is not fully defined.
- **Strongly recommended in TB meningitis** (mortality benefit). Sometimes used in severe pulmonary TB with **paradoxical worsening** on treatment, and in pericarditis to prevent constriction.
- In UG TB, **use to prevent ureteric strictures and bladder contraction is anecdotal only** (no trials). A meta-analysis across pulmonary, meningeal, pleural, pericardial and peritoneal TB found that **steroids reduced mortality whichever organ was affected**.

Surgical Therapy

- **About 55% of UG TB patients need surgery** at some point, more often as disease advances.
- **Indications:**
 - relieve obstruction or drain infection;
 - remove a nonfunctioning infected kidney resisting cure;
 - treat medically resistant hypertension from a functionally excluded kidney;
 - reconstruct the urinary tract (**now over half of TB operations**);
 - drain unresponsive abscesses by aspiration, percutaneous or open drainage.
- **Optimal timing: 4–6 weeks after starting medical therapy**, so that inflammation settles, bacillary load falls and lesions stabilise.

Procedures to relieve obstruction. Emergency drainage for uraemia or sepsis. Renal failure usually reflects **bilateral obstruction or obstruction of a functionally solitary kidney**. Early stenting or **percutaneous nephrostomy (PCN)** limits loss of renal function and preserves reconstructive options. Stepwise approach:

- **retrograde double-J stent (preferred)**, which succeeds in only about 41% → **antegrade** internalised or externalised stent via percutaneous puncture → **PCN** until definitive management (more than one may be needed because of strictures) → nephrectomy if unsalvageable.

- **PCN must be followed by correction of the cause:** removing it alone risks a **tuberculous cutaneous fistula** (less likely under effective therapy).
- **Avoid high contrast-injection pressures**, which risk dissemination.

Nephrectomy. Organ preservation is the fundamental goal.

Procedure	Indication	Outcome
Partial nephrectomy	Localised lesion (e.g., polar mass) failing medical treatment	—
Total nephrectomy (1)	Nonfunctioning kidney + recalcitrant or recurrent TB despite optimal therapy	Relapse <1% after short-course therapy
Total nephrectomy (2)	Nonfunctioning kidney + medically resistant hypertension	Hypertension improves in about 65%
Nephrectomy for nonfunction alone	Advocated by some	No supporting evidence; the bacilli-in-specimen studies predate modern therapy

- Nephrectomy is performed in about 27% of UG TB, similar in developed and developing countries.
- **Technique: oblique retroperitoneal incision** (extendable); remove diseased perinephric fat en bloc; **ligate artery and vein separately** (late arteriovenous fistula); **ureter usually left**; spare lymphatics, pleura and peritoneum.
- **Laparoscopic nephrectomy** works despite fibrosis, with less blood loss and faster recovery, and some advocate it as the preferred approach. It takes only modestly longer than for other indications.

Ureteropelvic and ureteral surgery. Stent PUJ and ureteric strictures temporarily to let renal function recover.

Upper and mid-ureteric strictures are rare and may suit endourology; **lower strictures are commoner and often need open surgery.** Decision factors: **stricture length and degree, whether a guidewire passes, blood supply and renal function.**

Endoscopic management. TB strictures have **mucosal ischaemia and dense fibrosis**, so success rates from other aetiologies do not transfer. **Best results: short strictures with a residual lumen and good renal function.** Strictures forming on treatment and **stented early may stabilise** without further intervention. Retrograde or antegrade **balloon dilatation** of ureter, PUJ, vesicoureteric junction or infundibulum is usually followed by a stent; **failure is high and repeat procedures common.**

- **Image all ureteric strictures during follow-up** (US or IVU), especially endoscopically treated ones, because healing fibrosis may worsen them. **Add steroids if they deteriorate.**
- **Failure to improve or progression after 6 weeks of medical therapy → open surgery.**

Open surgical options. Long, complex strictures need open repair, made harder by fibrosis, inelasticity and poor vascularity. **PUJ repair is harder in TB than in congenital stenosis.**

COMPARISON: RECONSTRUCTIVE OPTIONS BY LEVEL

Level / problem	Procedure	Notes
PUJ, short segment, extrarenal pelvis	Dismembered pyeloplasty	—
PUJ, longer stricture	Nondismembered (flap) pyeloplasty	May be impossible if pelvis heavily scarred
PUJ, reconstruction impossible	Ureterocalicostomy (ureter to lower-pole calyx)	Preserve capsule to cover lower pole; omentum if capsule insufficient
Upper / mid ureter	Excision + tension-free ureteroureterostomy ; or Davis intubated ureterotomy	Adequate mobilisation essential

Level / problem	Procedure	Notes
Lower ureter	Excise to healthy, well-vascularised mucosa + ureteroneocystostomy	Bridge gap to bladder as below

COMPARISON: GAP-BRIDGING MANOEUVRES FOR LOWER URETERIC RECONSTRUCTION

Manoeuvre	Gap bridged	Caveat
Contralateral bladder mobilisation ± dividing superior vesical artery	2–3 cm	—
Psoas hitch	Up to 5 cm	Needs good capacity; avoid genitofemoral and femoral nerves
Boari flap (± psoas hitch)	10–15 cm	Poor technique reduces capacity; a contracted TB bladder may lack area and elasticity
Ileal ureter (interposition)	Whole ureter	Multiple or recurrent strictures; native ureter no longer adequate

Bladder surgery.

Augmentation cystoplasty	Orthotopic bladder substitution
Intolerable frequency, nocturia, urgency, pain, haematuria, typically capacity <100 mL . First described in the 19th century for TB.	“Thimble” bladder <20 mL
Severe contraction → ileocaecum or sigmoid ; half the bladder diseased → ileum ; stomach and caecum also used	—

- General rules for bowel in the urinary tract apply: assess renal function, **reconfigure to a low-pressure reservoir**, educate the patient and follow up long term. Complications: **mucus, electrolyte derangement, bacterial infection**. Laparoscopic and robot-assisted augmentation is increasingly used.

Urethral procedures. **Bladder neck contracture → transurethral incision.** Urethral strictures are managed endoscopically and often recur. **Tuberculous urethral fistula → medical therapy + suprapubic drainage, then delayed reconstruction.** A seminal vesicle cavity has been drained into the bladder by cold-knife incision.

Genital surgery. Only after medical failure. Sequence: **epididymectomy alone if the testis is spared** (preserve testicular blood supply; ligate the vas, then start dissection at the **globus minor**) → **scrotal orchidectomy if the testis is involved**. Vasal disease is usually **distal to the external ring**, so ligation at the ring suffices.

Monitoring for Tuberculosis Relapse

	Pulmonary TB	UG TB
Relapse rate	2–6% (mostly in year 1)	6.3–22%, even after 12 months of therapy
Follow-up	2 years	Up to 10 years suggested (mean time to relapse 5.3 years)

- **Why UG TB relapses more:** the diseased kidney holds **innumerable foci that are hard to sterilise**; viable bacilli have been found after 9 months of therapy.
- Relapse needs a **longer or different course**; always **re-isolate the organism for susceptibility testing**.

Concept — Treatment is *medical first and surgical second*, and the timing follows the biology. Drugs sterilise but healing scars, so obstruction can worsen on treatment. Surgery waits 4–6 weeks for the bacillary load to fall, drains early to save nephrons, and reconstructs once fibrosis has declared itself. Surveillance runs for years because the kidney hides foci.

Management of Urogenital Tuberculosis in Special Situations

- Each situation needs **careful regimen selection for side effects, interactions and toxicity**, with **expert infectious diseases input**.

Multidrug-Resistant, Extensively Drug-Resistant, and Pre-Extensively Drug-Resistant Tuberculosis

COMPARISON: DRUG-RESISTANCE DEFINITIONS (US, 2022)

Category	Resistant to	Treatment and cure
MDR-TB	INH + rifampicin	Conventional regimens >18 months , sometimes intramuscular drugs; cure 50–60% (vs 94–97% drug-susceptible)
Pre-XDR-TB	INH + rifampicin + fluoroquinolone OR injectable aminoglycoside	—
Extensively drug-resistant TB (XDR-TB)	INH + rifampicin + fluoroquinolone + ≥ 1 injectable (amikacin, kanamycin, capreomycin), or + bedaquiline or linezolid	5–6 drugs for ≥ 2 years; cure 30–50%

- MDR-TB occurs in about 3.6% of new and 20% of previously treated patients worldwide, and in **<1.5% of US cases** (stable for 20 years). About one-fifth of MDR-TB has additional resistance.
- **BPaL (bedaquiline + pretomanid + linezolid)**, since 2019: an **all-oral MDR-TB regimen of 6–9 months**, also used successfully in XDR-TB without bedaquiline or linezolid resistance. **Not approved for extrapulmonary TB; its role in UG TB is undetermined.**

Pregnancy and Lactation

- **Advise avoiding pregnancy during treatment.** If TB is found in pregnancy, **treat promptly: the fetal risk from TB exceeds the drug risk.**
- **Regimen: INH + rifampicin + ethambutol + pyridoxine for 9 months.** Pyrazinamide is controversial in the US (uncertain fetal effects, perceived hepatotoxicity), but **WHO recommends it.**
- **Breastfeeding is allowed once noninfectious:** breast-milk drug levels are too low to be toxic.

Human Immunodeficiency Virus Infection

- **HIV raises the risk of active TB 15- to 20-fold**, and each infection accelerates the other. **Test every TB patient for HIV.** TB causes >30% of deaths among PLHIV worldwide.
- Extrapulmonary and UG TB may be commoner in PLHIV, with more dissemination, lymphadenopathy and **bilateral renal disease**. Postmortem UG TB is frequent in AIDS.

UG TB with HIV	UG TB without HIV
Less caseation, necrosis and fibrosis (fibrosis needs a competent immune response)	Vigorous inflammation → scarring
Collecting-system stenosis uncommon ; bladder contracture uncommon	Stenosis and contracture common
High mortality despite fewer obstructive lesions	—

- **Do not delay treatment;** guidelines are as for HIV-negative patients. **6 months is effective** and 9 months is no longer routine. **Extend to 9 months if the patient is not on consistent, effective antiretrovirals.** **Daily (not intermittent) dosing** is preferred.
- **Rifamycins (rifampicin more than rifabutin) lower antiretroviral levels**, so antiretroviral doses may need increasing.

Renal Transplant Recipients

- **Allograft TB is rare**; it usually presents **within 6 months of transplantation** (range up to 7 years), reflecting immunosuppression or donor-derived TB.
- **Early presentation and immunosuppression mean imaging is usually normal** and typical UG features are absent. **Fever is the usual presenting symptom**; urinary symptoms occur in only about 20%. The chest radiograph is often abnormal but nonspecific; **sputum culture is often positive**, and **urine AFB culture was positive in all cases in one series**. Many are diagnosed only at **graft nephrectomy**.
- **Rifamycin-immunosuppressant interactions** require drug-level monitoring. **Rifamycin-free regimens extend treatment to ≥ 18 months**. Complications: rejection, dissemination and **death in up to 36%**.

Concept — Host immunity shapes the disease. Where immunity is blunted (HIV, transplant), there is less fibrosis and fewer strictures but more dissemination and death. The regimen problem shifts from sterilisation to **rifamycin interactions**.

KEY POINTS: TUBERCULOSIS

- Untreated UG TB causes irreparable damage (renal failure, infertility); consider TB in the differential of UG disorders.
- Symptoms are nonspecific, the course indolent and constitutional symptoms infrequent, so a high index of suspicion is needed.
- Gold standard: positive AFB culture of urine or tissue. NAATs or granulomas on histopathology often allow the diagnosis.
- Optimal timing of surgery is 4–6 weeks after starting medical therapy.
- Monitor for relapse after treatment; it occurs in up to 22%.

PARASITIC INFECTIONS OF THE UROGENITAL TRACT

- These are **rare in nonendemic practice**, but urologists must know them to diagnose and treat. The relevant infections are **UG schistosomiasis, filariasis, amoebiasis, enterobiasis and echinococcosis**.

Schistosomiasis

- **More than 200 million people are infected** with *Schistosoma*. UG schistosomiasis combines a complex life cycle, multifaceted disease and tight ecological links.

COMPARISON: THE THREE MEDICALLY IMPORTANT SCHISTOSOMES (WITH FIG. 31.8)

Species	Geography	Adult worm site	Eggs shed in	Snail host	Main disease
<i>S. haematobium</i>	Africa, Arabian Peninsula	Vesical (bladder) venous plexus	Urine	<i>Bulinus</i>	UG: haematuria, bacterial UTI, scarring, calcification, SCC
<i>S. mansoni</i>	Africa, Arabian Peninsula, South America	Mesenteric venules (bowel/rectum)	Stool	<i>Biomphalaria</i>	Liver/GI: presinusoidal granulomas, Symmers pipestem periportal fibrosis , portal hypertension
<i>S. japonicum</i>	China, Southeast Asia	Mesenteric venules	Stool	<i>Oncomelania</i>	As <i>S. mansoni</i> ; Katayama fever especially

Fig. 31.8 from King CH: Schistosomiasis. In: Tropical infectious diseases, 2006. *S. intercalatum* and *S. mekongi* are also intestinal. All species occasionally cause embolic egg granulomas in brain or spinal cord. UTI, urinary tract infection.

- ***S. haematobium* probably coevolved with humans and nonhuman primates**, so ancient civilisations already recognised the syndrome.

History

- Schistosome antigens are found in Egyptian mummies (c. 3500 BCE), and *S. haematobium* eggs in later ones. The Egyptians called it “**A-a-a disease**”, drawn as a **penis dripping bloody urine**.
- **Theodor Bilharz, Cairo, 1852**: found worms in the mesenteric veins and linked them to eggs in urine and stool.

Biology and Life Cycle

- **Human cycle**: cercariae penetrate **even intact skin** in infested fresh water (“swimmers’ itch”) → shed forked tail over 90–120 min → **schistosomulae** → blood or lymph to the **lungs** (several weeks) → **liver sinusoids** (begin blood feeding, mature) → mate → migrate to the **vesical and pelvic venous plexus**. Cercaria to adult takes **80–110 days**.
- **Pairing**: the male clasps the female in the **gynaecophoric canal**, helps her feed and chemically stimulates egg-laying.
- **Eggs**: hundreds to thousands per female per day; *S. haematobium* eggs are **ovoid, about 140 µm, with a terminal spine** (Fig. 31.9). They must cross the endothelium and bladder wall into urine; some pass in faeces. **Fewer than half are excreted; retained eggs drive the pathology**.
- **Environmental cycle**: eggs survive about 20 days if wet → hatch in fresh water and light → **ciliated miracidia (lifespan ~6 h) penetrate the snail** (*Bulinus* tolerates slow, low-oxygen water) → primary sporocyst → 20–40 daughter sporocysts → **200–400 cercariae each** → released into water. **No cercarial shedding below 15 °C or above 35 °C**.
- **Cercarial lifespan and infectivity**: average about 1 day in the text (1–4 days in Fig. 31.8); penetration success falls within hours of shedding.
- **Adult lifespan**: average **3–5 years** (Fig. 31.8: 3–30 years; survival for years to decades).

Epidemiology

- **Endemic across sub-Saharan Africa, parts of North Africa and the Middle East**, limited by the tropical needs of the parasite and its snails.
- **Corsica, 2013**: the **first autochthonous European transmission in over a century**, with further cases in 2015 suggesting an endemic focus. Some infections were due to *S. bovis* or *S. haematobium*–*S. bovis* hybrids.
- Most of the >200 million (up to 440 million on newer estimates) have the UG form; **up to 150,000 deaths a year from obstructive renal failure alone**.
- **Risk depends on the nature and duration of freshwater contact**: domestic chores (rural women and children), play and swimming, fishing, car-washing, and water-intensive agriculture (rice, sugar cane). Prevalence is **highly focal**.
- **Dams and irrigation increase transmission**: they create year-round, slow-flowing snail habitat and denser farming populations. After the **Aswan Low Dam (1930s)**, prevalence rose from <11% to >75% in places; the **Aswan High Dam (1960s)** and **Diana Dam (1980s)** did the same.
- Prevalence is highest where there is **no clean running water or sanitation**. **Peak burden is at age 5–15** (whether from exposure or susceptibility is unclear).
- **Link with poverty**:
 - snail hosts live in tropical developing countries;
 - inadequate sanitation lets eggs reach water;
 - lack of safe water prolongs contact;
 - infection itself impairs growth, learning and productivity, perpetuating poverty.

Pathogenesis and Pathology

- **Cercarial penetration** uses proteases but provokes little beyond local skin inflammation (early immunomodulation). **Repeated exposure → hypersensitisation and maculopapular rash.**
- **The tegument** is a double lipid bilayer that permits years of survival. Immune evasion uses host antigen mimicry, membrane turnover, immunomodulatory proteins and proteases, and altered surface antigen expression.
- **Disease is driven by eggs, not worms.** Oviposition in bladder and pelvic organs causes granulomas (eosinophils, plasma cells, lymphocytes) and fibrosis. **All granuloma stages coexist**; clustered eggs form composite granulomas. These appear as **bulky, hyperaemic polypoid masses** (Fig. 31.10: IVU scalloping of bladder and lower ureter by polyps).
- Disease severity is also modified by host genetics, in utero sensitisation and coinfection. **Eggs induce type 2 inflammation genes and suppress urothelial barrier genes**: host and parasite “share” the goal of expelling eggs across a temporarily weakened urothelium.
- **Ectopic egg deposition**:
 - portal: presinusoidal obstruction → portal hypertension; **pipestem (Symmers) fibrosis** → hepatosplenomegaly;
 - pulmonary: only in very severe infection → pulmonary fibrosis, pulmonary hypertension, cor pulmonale.

Resistant individuals	Chronically infected individuals
Mixed T-helper 1 (Th1) + Th2 response; stay egg-negative ≥5 years without treatment despite exposure	Th2 only ; protective immunoglobulin E (IgE) may be blocked by IgG4

- **IgE to worm antigens rises with age**, suggesting acquired resistance. Water contact also falls with age, but **age appears to matter even after controlling for exposure.**
- **Obstructive sequence**: unexpelled eggs **calcify** → reduced urinary tract compliance and raised upper-tract pressure → stasis, **hydroureter**, then **hydronephrosis**. The extent of calcification on imaging roughly tracks egg burden.
- **Level of obstruction**, commonest first: **juxtavesical ureter** (20–60%), lower third (15–50%), intramural ureter (10–30%), meatus (1%). A contiguous combination occurs in 30–60%.

COMPARISON: THE THREE PATTERNS OF SCHISTOSOMAL HYDROURETER

	Segmental (cylindrical / fusiform)	Tonic	Atonic
Share of obstructive uropathy	About one-quarter (80% lower ureter)	Up to 30%	Remainder
Wall	Dilatation above concentric fibrosis and sandy patches	Thick, trabeculated, muscular hypertrophy , tortuous	Thin, atrophic, fibrotic muscle , very tortuous
Peristalsis	—	Impaired; whole ureter above lesion = functional stenosis	Absent
Hydronephrosis	Rarely significant	Significant but reversible on relief of obstruction	—

- **Hydroureter precedes hydronephrosis.** Untreated, hydronephrosis progresses as: pelvic dilatation → medullary atrophy → medullary effacement → cortical atrophy. This is why **tubular function (concentrating ability) fails before glomerular function.**
- **Bacterial UTI is increased**: bacteria adhere to the worm tegument, and stasis and immunomodulation contribute.

- **Bladder cancer, especially SCC: the strongest helminth–cancer association**, supported by Egyptian epidemiology and experimental models. Risk rises with duration and severity; mortality reaches about 10.8 per 100,000 Egyptian men.

Female genital schistosomiasis (FGS)	Male genital schistosomiasis (MGS)
Eggs in uterus, cervix, lower genital tract → fibrotic nodules, “ sandy patches ” (Fig. 31.11); increased mucosal vascularity	Ectopic eggs in testis, epididymis, cord, scrotum, penis (and prostate: higher prostate-specific antigen, PSA)
Poorly understood	Ill-characterised; internal organs need transrectal US
Conjectured to increase HIV susceptibility	Same conjecture
—	Genital disfigurement, haematospermia , haematuria, infertility (reversible) , from epididymitis and funiculitis; azoospermia with eosinophilic testicular infiltrate), pain; sexual dysfunction; urethral discharge mistaken for STI; fibrosis, calcification, neoplastic growth
—	Diagnosis: eggs in ejaculate. Symptoms predict poorly; semen egg counts vary and correlate poorly with urine; effect of abstinence unknown. Hydrocele and testicular calcification may be incidental. Lower serum testosterone (mechanism unknown). Treat with praziquantel .

- **Carcinogenic sequence:** urothelial hyperplasia → squamous metaplasia → dysplasia → urothelial carcinoma or SCC. **Bladder cancer is the final sequela.**

SCHISTOSOMAL BLADDER CANCER

Feature	Detail
Onset	Early: 40–50 years
Histology	SCC 60–90% ; adenocarcinoma 5–15%; >40% of SCCs well differentiated and verrucous, with good prognosis
Site	Posterior wall about half ; lateral wall about 30%
Growth	Exophytic two-thirds ; ulcerative endophytic one-third
Effect of mass drug administration (MDA), Egypt	Bladder neoplasms fell from 28% to 12% , with a shift from SCC to transitional cell carcinoma
Cofactors	Smoking (possibly synergistic); some unselected autopsy series found similar cancer rates without schistosomiasis
Classification	International Agency for Research on Cancer (IARC) Group 1 (“carcinogenic to humans”)

• Mechanisms:

- egg deposition in the bladder wall;
- early rise in **vascular endothelial growth factor (VEGF)**, promoting tumour vasculogenesis;
- papillomas merging with basal epithelium → fibroepithelial growths, then repeated inflammation and fibrosis → clonal precancerous foci (squamous metaplasia);
- transcriptional changes in carcinogenesis pathways (mouse bladder);
- **bacterial coinfection reduces nitrates to nitrosamines** → protein and DNA alkylation → mutations (e.g., p53).

Fig. 31.11 — *macroscopic stages of urinary schistosomiasis*. (A) **Early active:** erythematous granular sessile and pedunculated polyps on posterior and apical walls. (B) **Chronic inactive:** yellow-tan granular **sandy patch** replacing lamina propria, with sandy foci through atrophic detrusor and perivesical fat; superficial viable eggs

with granulomas mark **chronic active** disease. (C) Concentric sandy patch at the margin of an exophytic SCC, with no recent oviposition: the chronic inactive stage typical of the **bilharzial bladder cancer syndrome**. (D) Severe chronic inactive disease with epidermisation, a **dilated “golf-hole” ureteric orifice** on one side and a stenotic orifice on the other. (E) Rectosigmoid polyposis with active oviposition. (F) **Stellate chronic schistosomal ulcer** in a partial cystectomy specimen from inactive disease.

Concept — Worms are silent; eggs cause disease. Every lesion — polyp, sandy patch, stricture, calcification, cancer — comes from retained eggs and the host granuloma. That is why pathology scales with egg burden and duration, and why early lesions reverse while calcified fibrosis does not.

Clinical Manifestations

- **Acute** disease is the response to penetration and migration. **Chronic** disease is the immune response to years of oviposition, so it is **confined to long-term residents** with repeated reinfection and heavy burdens.
- **Schistosomes cannot replicate in humans**, so a single-exposure traveller’s worms **die of senescence within 3–5 years**, limiting pathology even without treatment.

Acute schistosomiasis	Chronic schistosomiasis
Cercarial dermatitis: itchy maculopapular rash 1–2 days after exposure, often gone before the traveller returns	Oviposition into bladder from 8–12 weeks ; painless recurrent haematuria , dysuria, frequency; proteinuria
Katayama fever 2–8 weeks later (often silent): fever, dry cough, fatigue, headache, diarrhoea, eosinophilia , neck pain, urticaria	Haematuria is specific enough to be the primary screening sign in endemic areas ; seen as a “sign of puberty” in boys in some cultures and can cause anaemia; often missed in returning travellers
Katayama fever: commonest with heavy primary <i>S. japonicum</i> , less with <i>S. mansoni</i> , rare with <i>S. haematobium</i> ; rare in endemic residents; confused with malaria or enteric fever	Fibrosis → obstruction, hydroureteronephrosis, vesicoureteric reflux . Anthelmintics may restore renal function in shorter infections; surgery or diversion in late disease

- Because acute disease may be silent, **anyone exposed to potentially infested water should be considered for testing and treatment.**
- **FGS** affects 33–75% of infected women (eggs in tubes, cervix, vagina, vulva, ovaries, uterus). **Friable sandy patches bleed on contact.** Dyspareunia, pelvic pain, bleeding, discharge, frequency and infertility are **misdiagnosed as UTI or STI.**
- **Men:** heavy egg loads in ejaculatory ducts and seminal vesicles; **blood or eggs can appear in ejaculate before urine.** Presentation can be a testicular mass or scrotal pain. **Egg burdens are higher in epididymis, ovaries and tubes than in testis, uterus and vagina.**
- **Late chronic active stage** (peak tissue egg burden) produces the “**schistosomal contracted bladder**”: suprapubic and pelvic pain, urgency, frequency, incontinence. The trigone looks normal or mildly hyperaemic while the rest of the wall is thick and indurated; **functional capacity can fall to 50 mL.**
- **Quiescence over years:** oviposition falls and many light infections become asymptomatic, **but silent obstructive uropathy progresses** as fibrosis replaces polyps, so severe hydroureteronephrosis can develop insidiously.
- **Chronic inactive stage:** no viable eggs. Symptoms come from reactions to calcified dead eggs. **40–60% of schistosomal obstructive uropathy presents to urologists at this end stage.** Nonfunctioning kidneys in asymptomatic people are common in heavily endemic areas.
- **Bacterial UTI complicates about half of obstructive uropathy**, with the usual organisms.

- **Salmonella association:** chronic or recurrent *Salmonella* UTI with intermittent bacteraemia (bacteriuria as “spill-over”). The bacteria shelter in **tegument invaginations**, so **treat both infections; antibiotics alone fail**.
- **Bladder ulcers:** **acute ulcers are rare** (necrotic polyps slough in active disease). The **chronic ulcer is commoner**, a late sequela of heavy infection with constant burning, intense pain, gross haematuria and pyuria.
- **Eosinophilia:** very common acutely and low-grade chronically; neither sensitive nor specific, but a clue.
- **Exceptions to the usual sequence:** ectopic pulmonary schistosomiasis, neuroschistosomiasis and FGS.

Diagnosis

COMPARISON: DIAGNOSTIC TESTS FOR UG SCHISTOSOMIASIS

Test	Key facts	Limitation
Eggs in urine or stool (gold standard, active infection)	Collect urine 09:00–15:00 (shedding peaks at noon); concentrate for light infections	Eggs appear only from 8–12 weeks
Tissue biopsy	Rectal snip before bladder biopsy (eggs common in rectum; avoids bladder infection or perforation). Squash preparation beats histology (more sensitive, shows viability). Vulval, vaginal or cervical biopsy for FGS	Invasive
Haematuria and turbidity (visual, dipstick)	Used in control and elimination campaigns	Less sensitive and specific
Serology (FAST-ELISA + Western blot, CDC)	>90% sensitive and specific ; positive from 4–6 weeks	Cannot separate acute, chronic or cured infection (stays positive after cure)
Circulating anodic / cathodic antigens (CAA, CCA)	Released only by live adult worms , so specific for active infection and quantitative; point-of-care urine CCA strip	CCA has failed to detect <i>S. haematobium</i> in some hands (performs better for <i>S. mansoni</i>); expensive
PCR	Most sensitive (urine, stool; vaginal lavage for FGS)	Needs trained staff, solvents, kits; impractical in the field. Useful for surveillance in low-transmission settings

- **Imaging:**
 - **US:** wall thickening, polyps, hydroureter, hydronephrosis, calcification, even carcinoma.
 - **Plain film:** calcification; the **calcified bladder resembles a fetal head in the pelvis** (Fig. 31.12: rim of calcification on plain film; bright line with a dark rim behind it on US). Calcification also in prostate, seminal vesicles, posterior urethra, distal ureters, occasionally colon.
 - **IVU:** **earliest sign is striation** of ureters and pelvis. Also hydronephrosis, hydroureter, nonfunction, stenosis and polypoid filling defects. Needs delayed films in severe obstruction; post-void films show bladder neck obstruction. **With fluoroscopy it distinguishes tonic from atonic ureters.**
 - **CT:** obstruction plus colonic and urinary calcification (advantage over IVU). **MRI:** no clear advantage yet.
 - **Voiding cystourethrography:** **reflux in 25% of infected ureters.** Cystourethroscopy shows papillary and submucosal lesions (Fig. 31.13, a Somali child with haematuria); retrograde pyelography shows ureteric anatomy.

Ureteric calcification in schistosomiasis	Ureteric calcification in TB
Intramural, in a dilated ureter	Intraluminal casts of a nondilated ureter

- Globally, **most cases go undetected because diagnosis relies on direct egg detection**. Better tools are central to control.

Treatment

Medical management.

- **Praziquantel (PZQ) is the only WHO-recommended drug**. Single-drug dependence risks resistance, but PZQ is effective, available and low in toxicity.
- **Dose for *S. haematobium*: 20 mg/kg orally twice on the same day, 6–8 h apart (or 40 mg/kg once)**. Add corticosteroids for Katayama fever.
- **Efficacy (egg reduction and cure) 60–90%**; uncured patients still have much lower worm burdens. Monitor with urine and stool egg counts and serial US; repeat courses if infection persists.
- **Side effects** (abdominal pain, nausea, headache, dizziness) are mild, occur within 3–4 h and resolve. Tablets are large and bitter (hard for children); a paediatric formulation is in development.
- **Pregnancy**: US Food and Drug Administration (FDA) Category B (animal data). Many programmes exclude pregnant and lactating women, **yet few adverse effects have been reported among millions treated**.
- **PZQ is less active against schistosomulae than adults**. It therefore **cannot abort infection soon after exposure**, which partly explains lower cure rates where transmission is high. **Repeat doses weeks apart** catch worms that have since matured.
- **Resistance** is debated: large endemic studies show little, but failures occur in returning travellers and military personnel. Environmental control may reduce reliance on PZQ.

	Praziquantel	Artemisinins (artemether, artesunate)
Target stage	Adult worms	Schistosomulae
Use	Treatment of established infection	Chemoprophylaxis (90–97% preventive); poor for established infection
Combination	PZQ + artemisinin lowers infection rates more than PZQ alone and is useful in high-transmission MDA, but it is not for widespread use where malaria is coendemic (risk of artemisinin resistance)	

Surgical management. Because PZQ is effective and **early disease is reversible, trial medical therapy before elective surgery**. Reserve surgery for complications unresponsive to adequate treatment, or emergencies, **commonly severe bladder haemorrhage**.

- **Bladder outlet**: prostatitis and enlargement are uncommon and autopsies show no anatomical outlet obstruction. However, cystoscopic, urodynamic and post-void residual data show **functional outlet obstruction**, occasionally needing surgery in severe inactive disease.
- **Scrotal induration, pain and enlargement from epididymitis** can lead to surgery for suspected testicular tumour.
- **Irreversibly contracted bladder** → vesical denervation, urinary diversion, ileocystoplasty or hydrodistension, **always with chemotherapy**.
- **Chronic deep ulcers** → **partial cystectomy** (fulguration rarely relieves symptoms or heals them). Hyperplasia accompanies severe disease; dysplasia and metaplasia accompany cancer. **Schistosomal bladder cancer is treated surgically** (Chapters 130–132).
- **Ureteric obstruction is the commonest sequela** and correlates with infection intensity. It is usually caused by **polypoid lesions girdling the intramural and adjacent ureter**, so it **often responds to medical therapy alone**. Renal function lost to active-infection obstruction **recovers within 1–2 months of PZQ**, and

chemotherapy also **prevents** obstruction despite reinfection. **Late chronic and inactive disease responds less well.**

- **Anatomical stenosis** ($\pm$ calculi) is found in up to 80% of obstructed ureters. Residual stenosis after chemotherapy is usually surgically correctable:
 - **balloon dilatation** works, but **mechanical dilatation often restenoses**;
 - meatus, intramural ureter, vesicoureteric junction or distal ureter → antireflux reimplantation, mostly **Politano-Leadbetter variants**. Effective, but restenosis occurs;
 - **long or multifocal lesions** → Boari flap, ileal conduit, **suprapubic intravesical ureterostomy** (peritoneal dialysis catheter bypass into bladder), or **ileal ureter kept isoperistaltic**;
 - **isolated meatal stenosis** → **meatoplasty**;
 - **unreconstructable** → **long-term nephrostomy**.

Prognosis

- **Most infections are mild, with a good prognosis.** Morbidity and mortality depend on **infection intensity and immune-gene polymorphisms**.
- Low-prevalence regions (e.g., Nigeria) have essentially no mortality. **When Egyptian prevalence was 50%, schistosomiasis contributed to death in 10% of those infected**; mortality in severe disease approached 50% over 2–5 years.

Death from obstructive uropathy	Pyelonephritis and urothelial cancer
20s, heavy egg burden; bilateral end-stage hydronephrosis (or unilateral with contralateral end-stage kidney)	Over 40 (time- and intensity-related)

- **PZQ has transformed prognosis:** obstructive polyps in children resolve within **2–6 weeks**. Prognosis is less certain for fibrotic, sandy-patch obstruction, though some tolerate it with stable function.
- Obstruction, stones, outlet obstruction and cystitis predispose to **pyelonephritis, which can be life-threatening: treat promptly and aggressively**. Cancer prognosis depends on tumour aggressiveness.

Prevention and Control

- **Travellers: avoid fresh water. Fast-flowing water can still be infested.** Heating to $\geq 125^\circ\text{F}$ (about 52°C) for 5 min, chlorination, or standing >2 days away from snails kills cercariae.
- **PZQ (late 1970s; MDA from the 1980s–90s)** made treatment simple and cheap, but control remains difficult. WHO strategy has been **periodic (typically annual) targeted MDA for morbidity control**. Without safe water, reinfection forces more frequent dosing.
- Cost is typically $<\text{US\$}0.50$ per person but still burdensome at scale; **donated PZQ** supports campaigns.
- **Asexual amplification in snails** means **egg input must fall by nearly 90%** before transmission falls substantially. Snail reduction lowers transmission proportionally, but adult worms persist for years, so **snail control alone must run for years**.
- **Integrated control** offers the most promise: treat people, **separate people and excreta from water**, and control snails. Other measures: molluscicides, biological control (snail predators and competitors), **WASH (water, sanitation and hygiene)**. There is no effective vaccine yet.
- Unlike malaria and lymphatic filariasis, **global prevalence has not fallen**. Nevertheless it has been **eliminated in 10 countries** (Iran, Japan, Lebanon, Malaysia, Martinique, Montserrat, Morocco, Thailand, Tunisia, Turkey).

- **WHA65.21 (65th World Health Assembly, May 2012):** marks the **shift from morbidity control to elimination**.

Concept — PZQ controls morbidity but not transmission, because the snail amplifies every egg that reaches water. Durable gains need water, sanitation and snail control alongside drugs.

KEY POINTS: SCHISTOSOMIASIS

- *S. haematobium* survives years to decades; a careful travel and social history identifies exposure and guides testing.
- PZQ in early UG schistosomiasis can reverse inflammatory lesions, including fibrosis, caused by the host response to eggs.
- Gold standard for active infection: eggs in urine, stool, or bladder or rectal biopsy. Serology and PCR are highly sensitive but may not distinguish active from resolved infection and are impractical in endemic regions.

Filariasis

- **Filariae are vector-borne tissue nematodes:** the agents of **lymphatic filariasis (LF)**, *Onchocerca volvulus* and *Loa loa*.
- **LF is caused by mosquito-borne *Wuchereria bancrofti*, *Brugia malayi* and *B. timori*.** It ranges from acute lymphatic inflammation to chronic dilatation with **hydrocele, lymphoedema and elephantiasis**.

Organisms

- **Lifecycle: infective third-stage larvae** enter by mosquito bite → migrate to central lymphatics → mature over **6–9 months** into adults (about 20–100 mm × 0.2 mm; microfilariae about 200 × 10 µm, Fig. 31.14). Adults live about **5–7 years**.
- **Adult site:** afferent lymphatics, especially of the lower limb (inguinal, iliac, para-aortic), and, **for *W. bancrofti* only**, male genitalia (epididymis, cord, testis).
- **Microfilaraemia peaks around midnight** in most areas, matching vector biting; in parts of the Pacific *W. bancrofti* periodicity is diurnal. In the mosquito, microfilariae become infective larvae in 10–14 days.
- ***Wolbachia*:** an obligate endosymbiont needed for embryogenesis. **Doxycycline kills it**, reducing microfilarial release and larval moulting.

Epidemiology

- **About 68 million infected**, much reduced since the early 2000s. ***W. bancrofti* causes >90%:** sub-Saharan Africa, South and Southeast Asia, western Pacific; in the Americas only Haiti, Dominican Republic, Guyana, Brazil. *B. malayi*: Asia and Pacific islands. *B. timori*: southeastern Indonesia only.
- **Vectors:** *Anopheles* (rural Africa, Pacific), *Culex* (urban, especially India), *Aedes aegypti* (some Pacific islands).
- **Transmission is inefficient**, so obstructive disease is seen mainly in **long-term residents with years of reinfection**. Prevalence rises to the third or fourth decade, then plateaus. **Lymphoedema and genital disease are rare before age 10**. About one-third of infected people have overt disease; clinical disease is likelier in India, Papua New Guinea and Africa than in the Americas.

Pathology and Clinical Manifestations

- **Early response is proinflammatory** (Th1 and Th2, rising filaria-specific IgE, eosinophil killing of microfilariae). **With microfilaraemia, T-cell responses fall** (IL-10, IgG4 blocking antibodies, suppressor T cells). Some people stay uninfected despite long exposure.

- **Damage is cumulative** (scarring, lymphatic obstruction). **Drugs prevent progression but do not readily reverse damage.** LF is rarely fatal but **third among parasitic infections for disability-adjusted life years (DALYs) lost.**
- **Lymphoedema cycle:** parasite factors cause the initial lymphatic dilatation; secondary bacterial infection, reactions to dying worms and *Wolbachia* add to it. Lesions range from nodular inflammation to suppuration (granulomas around worms $\pm$ eosinophils; Fig. 31.15). The vicious cycle runs: acute attack $\rightarrow$ worse lymphoedema $\rightarrow$ more infection $\rightarrow$ worse lymphoedema. The end state is **obliterated, scarred lymphatics $\rightarrow$ elephantiasis or hydrocele.**

Subclinical infection. Most are clinically silent even with heavy microfilaraemia, yet **virtually all patent infections have subclinical disease** (dilated lymphatics, scrotal lymphangiectasia, microscopic haematuria, proteinuria). **Eosinophilia is common.**

Acute adenolymphangitis (ADL). Often the first manifestation: fever, lymphadenitis, lymphangitis and oedema lasting days to a week. **Lymphangitis is retrograde (spreads peripherally), unlike bacterial lymphangitis.** Limbs are involved in both bancroftian and brugian disease, but **genital lymphatics almost only in *W. bancrofti*:** funiculitis, epididymitis, scrotal pain, **lymph scrotum** (ruptured scrotal lymph vesicles, whitish discharge, secondary infection).

Dermatolymphangioadenitis (DLA). Fever, chills, myalgia and headache, with oedematous inflammatory plaques, hyperpigmentation, vesicles and ulcers at a skin-injury site. It **spreads proximally and is thought to be bacterial.**

ADL	DLA
Retrograde (distal) spread; filarial	Proximal spread; bacterial

Lymphoedema. **The commonest chronic manifestation**, lower limb more than upper; often bilateral but **asymmetrical**; skin may weep; breasts can be involved in women.

<i>W. bancrofti</i>	<i>Brugia</i>
Entire limb	Below the knee
Male genital disease very common	Genital disease uncommon

Genitourinary manifestations. Female genital involvement is poorly defined and probably uncommon. Genital disease usually begins at least in the teens. **Acute (usually unilateral) epididymitis or funiculitis with fever and malaise** lasting days is one of the commonest consequences of bancroftian filariasis.

Funiculoepididymitis. Palpable **cord-like swellings** and oedema. Usually self-limiting, but recurrence and chronic lymphoedema are common. **Sterility and orchitis are rare because the testis is usually spared.** It is **often mistaken for malignancy**, leading to surgery including orchidectomy. Varicocele can add pain and swelling. **Bacterial superinfection is rare but severe** (exquisite pain, septic thrombophlebitis).

Hydroceles. The chronic male genital lesion, sometimes huge (Fig. 31.16). Filarial and nonfilarial hydroceles are hard to tell apart and **parasites are rarely found in the fluid.** **Pointers to LF:** hydrocele with **cord or epididymal nodules** plus endemic exposure; a **thick fibrous tunica with cholesterol or calcium.** Usually painless unless complicated. Scrotal skin may be thick and brawny with oozing lymph. **Superinfection is rare with hydrocele alone but common with elephantiasis and lymph scrotum.**

Scrotal and penile elephantiasis. Mild scrotal oedema is common early; **penile oedema is unusual.** Massive enlargement is **late**, mostly where care is poor. **Genital elephantiasis rarely has any cause other than LF.**

Chyluria. Damaged GU lymphatics leak lymph into urine $\rightarrow$ **massive fat and protein loss** and possible malnutrition. It occurs **earlier than genital elephantiasis**, is **intermittent** and may remit.

Tropical pulmonary eosinophilia (TPE). An **allergic response to microfilarial antigens** (South and Southeast Asia): **nocturnal paroxysmal cough and wheeze**, fever, adenopathy, **high-grade eosinophilia**, **raised IgE**. Chest film ranges from normal to reticulonodular; lung function is restrictive ($\pm$ obstructive); biopsy shows eosinophilic interstitial pneumonitis. **Microfilaraemia is usually absent**.

Diagnosis

- In endemic residents, lymphoedema and male genital disease are **more likely to be LF** if other causes of oedema are absent. **Differential for funiculoepididymitis**: TB, *S. haematobium*, gonorrhoea. Nonfilarial hydrocele is common everywhere, but **filarial hydrocele occurs earlier and more often** in endemic areas.

Method	Key facts	Limitation
Adult worms in tissue	Definitive	Insensitive; needs surgery or biopsy
US of lymphatics — “filarial dance sign”	Live worms move distinctively; $\geq 80\%$ sensitive in some settings	—
Plain radiograph	Calcification suggestive in context	—
Giemsa-stained blood smear	Microfilariae; time by local periodicity (midnight for most)	Microfilaraemia in only a minority
Circulating antigen (W. bancrofti)	ELISA or point-of-care immunochromatographic card test (ICT) ; the newer Alere Filariasis Test Strip outperforms the BinaxNOW ICT	No antigen test for Brugia
PCR (blood)	Very sensitive	Not widely available
Antibody	IgG4 and species-specific antigens improve specificity; dipstick for <i>Brugia</i> . Usually positive in “burned-out” disease	Traditionally poor specificity
Lymphoscintigraphy	Reliably shows lymphatic damage	Does not distinguish LF from other lymphatic disease

- “Burned-out” LF (treated, or long after leaving an endemic area): **negative microfilaria and antigen tests do not exclude LF**.

Treatment

- Treat all microfilaraemic patients, symptomatic or not**, because nearly all have subclinical disease.

COMPARISON: ANTIFILARIAL DRUGS — ACTIVITY AND DOSING

Drug	Dose (LF unless stated)	Microfilaricidal	Macrofilaricidal (adult)	Role
Diethylcarbamazine (DEC)	2 mg/kg orally three times daily; 1 day as effective as 12 days ; TPE 2–3 weeks. Loiasis: 2–3 mg/kg three times daily for 14–21 days	Yes	Modest	Treatment of choice for active LF ; in the US via CDC only
Albendazole	400 mg twice daily for 21 days (LF); 200 mg twice daily for 3 weeks (loiasis)	Yes	Yes	Alternative; few side effects as single dose
Ivermectin	150–400 $\mu\text{g}/\text{kg}$ once (LF); 150 $\mu\text{g}/\text{kg}$ every 6–12 months (onchocerciasis)	Mostly	Some (repeat doses shorten adult lifespan by about half or more)	Alternative for LF; treatment of choice for onchocerciasis

Drug	Dose (LF unless stated)	Microfilaricidal	Macrofilaricidal (adult)	Role
Doxycycline	200 mg daily; 4–8 weeks (6 weeks in US practice)	Augments suppression	Some; sterilises adults (anti- <i>Wolbachia</i>)	Improves lymphoedema and hydrocele even without active infection

COMPARISON: ANTIFILARIAL DRUGS — TOXICITIES AND CAUTIONS

Drug	Adverse effects	Contraindication / caution
DEC	Fever, chills, arthralgia, myalgia, headache, nausea, vomiting. Reactions to dying worms or <i>Wolbachia</i>: painful nodules, lymphadenitis, epididymitis (days to weeks)	Not where onchocerciasis or <i>L. loa</i> are coendemic (West and Central Africa) unless excluded. Never in onchocerciasis (blindness, systemic toxicity)
Ivermectin	Similar to DEC in LF; in onchocerciasis fever, rash, dizziness, pruritus, myalgia, arthralgia, lymphadenopathy	Caution if <i>L. loa</i> possible (encephalopathy)
Albendazole	Few (single dose)	Lower encephalopathy risk than DEC in high-grade <i>Loa</i> parasitaemia
Doxycycline	—	Not in pregnancy or young children; long course impractical in developing countries

- **Chronic lymphoedema:** the basis of morbidity control is **preventing secondary bacterial infection, hygiene, elastic stockings, elevation and physiotherapy. Antiparasitic drugs only if infection is active.**
- **Surgery:**
 - challenging and often unnecessary. Lymphaticovenous and nodovenous anastomoses help limbs somewhat, and genital reconstruction can help; long-term results are unknown;
 - **genital elephantiasis is rarely amenable to surgery**, and **lymphadenectomy worsens drainage**;
 - **funiculoepididymitis:** decompression or excision of nodules to save testis and cord; **radical surgery if recurrent, painful, deforming or vascular.**
- **Hydrocele:** aspiration gives relief but recurs. **Hydrocelectomy for large or symptomatic hydroceles:** excision of the intact sac is the procedure of choice (inversion with partial excision is an alternative). **Suture or excise leaking or dilated lymphatics.** Small stable hydroceles need no surgery. Scrotal or vulval reduction helps selected patients.

Prevention and Control

- **Personal protection and long-lasting insecticide-treated bed nets (LLINs)** are valuable for control and elimination.
- **Transmission needs patent microfilaraemia**, so clearing microfilariae interrupts it. Drugs do not kill all adults, so **MDA must continue for years until adults die of senescence.** This works for *W. bancrofti* but is **harder for *Brugia* (animal reservoirs).**
- **MDA** gives single annual doses: **albendazole + ivermectin in Africa; albendazole + DEC elsewhere.** It is the main reason for falling prevalence and has **prevented about 19 million hydroceles since 2000**, halving LF hydroceles. **Triple therapy (DEC + albendazole + ivermectin)** suppresses microfilariae more deeply and for longer, without extra serious adverse effects (small studies).

Onchocerciasis (river blindness).

- **99% of cases are in Africa**, with foci in Brazil, Venezuela and Yemen. About 17 million are infected (down about 55% since 2000 through MDA). Transmission is inefficient and focal.

- **Adults live up to 15 years in subcutaneous nodules** (mean 9–10).
- **Clinical:** dermatitis and pruritus, keratitis, chorioretinitis → **blindness from corneal scarring**. **Urological relevance:** “**hanging groin**” and **scrotal elephantiasis** from recurrent lymphadenitis and loss of skin elasticity (atrophic, fibrotic inguinal nodes), and occasionally massive inguinal lymphadenopathy.
- **Diagnosis:** **skin snips**, adults in nodules, or **microfilariae in the anterior chamber on slit lamp**. Serology is less developed than for LF.
- ***O. lupi*** (a canine parasite) is a rare human pathogen: six US cases since 2013, mostly children in the Southwest, with nodules in spine, soft tissue and extraocular muscle.
- **Treatment:** **ivermectin 150 µg/kg every 6–12 months until asymptomatic**, then **6 weeks of doxycycline 200 mg daily**, which kills or sterilises most adult females. **DEC is contraindicated**.

Loiasis.

- **Central and West Africa;** adults migrate subcutaneously.
- Usually asymptomatic eosinophilia. Otherwise urticaria, migratory subcutaneous swellings and **eye worm** crossing the conjunctiva. **Haematuria and proteinuria in 30%**; lymphadenitis and hydrocele are rare.
- **DEC is effective** (may need several courses) but causes pruritus, arthralgia, swellings, fever, eye worm, diarrhoea and renal failure.
- **Microfilaraemia >2500–8000/mL → risk of treatment-associated encephalopathy** (especially with ivermectin). **Pretreatment apheresis** may help, and **albendazole is safer**.

COMPARISON: THE FILARIAE OF UROLOGICAL RELEVANCE

	<i>W. bancrofti</i>	<i>Brugia</i> spp.	<i>O. volvulus</i>	<i>L. loa</i>
Vector	Mosquito	Mosquito	<i>Simulium</i> black fly	<i>Chrysops</i> fly
Microfilarial periodicity / site	Nocturnal blood (diurnal in parts of Pacific)	Nocturnal blood	Skin and eye	Diurnal blood
<i>Wolbachia</i>	Yes	Yes	Yes	No
GU disease	Hydrocele, funiculoepididymitis, lymph scrotum, elephantiasis, chyluria	Genital disease uncommon	Hanging groin, scrotal elephantiasis	Haematuria and proteinuria (30%); rare hydrocele
Drug of choice	DEC	DEC	Ivermectin (DEC contraindicated)	DEC; albendazole if high microfilaraemia

Concept — Filarial disease is cumulative lymphatic injury. Drugs clear microfilariae and stop progression, but scarred lymphatics stay, so treatment of established disease is hygiene and surgery. Drug choice is dictated by **what dying parasites will provoke**: never DEC in onchocerciasis, and caution with ivermectin or DEC where *Loa* is present.

KEY POINTS: LYMPHATIC FILARIASIS

- LF is caused by three species of tissue nematodes transmitted by mosquitoes.
- LF is endemic across much of the tropics but transmitted inefficiently; clinical disease is usually confined to long-term residents.
- Diagnosis is by blood smear, antibody or antigen assays, and radiological and clinical means.
- Active LF is treated with DEC; ivermectin and albendazole are also active, and all three are used in mass chemotherapy programmes.

Other Nonfilarial Genitourinary Parasites

COMPARISON: OTHER UROGENITAL PARASITES

Parasite	Type / transmission	UG involvement	Treatment
Echinococcosis (<i>Echinococcus granulosus</i>)	Cestode ; ingestion of eggs in food or water, or contact with infected dogs; pastoral communities (South America, Mediterranean, Eastern Europe, Middle East, East Africa, Central Asia, China, Russia, Australia)	Cysts usually liver, then lung; kidney third (about 2–3%) . Grow 1–2 cm/year → pain or mass; hydatiduria and colic in a minority ; renal function usually preserved. Imaging: thick-walled spherical cyst, often calcified wall , which stages disease. Serology only adjunctive (60–90% sensitive). Rupture → anaphylaxis	PAIR (puncture, aspiration, injection, reaspiration) for liver cysts, not renal or lung. Renal cysts: surgical resection or albendazole 400 mg twice daily for 1–6 months. Perioperative PZQ + albendazole may reduce secondary seeding
Enterobiasis (<i>Enterobius vermicularis</i> , pinworm)	Worldwide (temperate and tropical); person-to-person faecal-oral via hands and fomites; worms in proximal colon lay perianal eggs, infectious after 6 h	Mostly asymptomatic; severe perianal pruritus . Rare ectopic migration via vagina, uterus and tubes to peritoneum → granulomas (vulval, cervical), salpingitis, oophoritis, tubo-ovarian abscess, appendicitis, peritonitis. Rarely epididymal and inguinal hernia involvement in males	Single-dose albendazole 400 mg or mebendazole 100 mg; alternative ivermectin 200 µg/kg. Treat household contacts; repeat at 2 weeks
Amoebiasis (<i>Entamoeba histolytica</i>)	Protozoan , faecal-oral, tropical	Mostly asymptomatic; 10% have symptoms in other organs including kidney. Cutaneous amoebiasis: painful perianal and genital ulcers	Tinidazole 2 g daily for 3–5 days or metronidazole 750 mg three times daily for 10 days, then a luminal agent: paromomycin 8–12 mg/kg three times daily for 7 days or iodoquinol 650 mg three times daily for 20 days
Trichomoniasis (<i>Trichomonas vaginalis</i>)	Common sexually transmitted protozoan	Covered in Chapter 29	

• **What this table tests:**

- **PAIR is a liver-cyst option only;**
- **amoebiasis needs a tissue agent then a luminal agent;**
- **pinworm needs contact treatment and a repeat dose**, because autoinfection and reinfection are the rule.